

VOTSPEECH: DECOUPLING VOICE DESIGN FROM SPEECH GENERATION IN INSTRUCTION-FOLLOWING TTS

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ABSTRACT

Instruction-following TTS often leaves voice design implicit in speech generation, as conventional speaker embeddings rely on reference audio rather than serving as explicit targets for text-driven design. We propose VoTSpeech (Voice-of-Thought Speech), which decouples voice design and speech generation through an explicit intermediate voice representation. Within a shared language-model backbone, VoTSpeech first predicts a continuous voice prior from text instructions via a flow-based module, and subsequently conditions acoustic decoding on this representation. To capture both speaker identity and fine-grained acoustic nuances, the representation is supervised by speaker identity features and audio VAE latents. Trained with modest task-specific adaptation data, VoTSpeech achieves state-of-the-art instruction-following accuracy on InstructTTSEval. Subjective evaluations further demonstrate superior expressiveness and instruction adherence alongside competitive naturalness, validating explicit voice priors as an effective interface for controllable speech synthesis. The audio samples are available at <https://ywbnn.github.io/VoTSpeech/>.

Index Terms— instruction-following TTS, voice design, decoupled voice modeling, flow matching.

1. INTRODUCTION

Natural-language voice design creates personalized voices from descriptions of vocal characteristics and speaking style without reference audio [1, 2]. While recent instruction-following models advance controllable speech synthesis [3, 4, 5, 6, 7], direct instruction conditioning remains a common approach. In this formulation, voice creation is implicit within utterance generation, without an explicit intermediate target for modeling the desired voice. This coupling obscures voice control and motivates separating *what voice to create* from *what that voice says*.

Explicit voice representations provide a natural bridge between voice descriptions and speech synthesis. While reference-conditioned TTS extracts speaker embeddings

from audio [8, 9, 10], voice design requires constructing a voice without such a reference. TacoSpawn takes a step in this direction by learning speaker embedding distributions [11], while PromptSpeaker and PromptTTS 2 further enable description-conditioned representation generation [12, 13]. For these representations to guide synthesis effectively, however, they must also carry useful acoustic information. This motivates complementary efforts, including synthesis-oriented speaker encoders in MiniMax-Speech [14] and hierarchical conditioning for timbre preservation and adaptive expression in CapTalk [15]. Together, these studies highlight the need to coordinate description matching with synthesis suitability. Effectively connecting text-driven voice creation with high-fidelity speech generation within a unified architecture therefore remains a non-trivial challenge.

To address this, we propose **VoTSpeech** (Voice-of-Thought Speech), which formulates voice design and speech generation as two connected tasks linked by an explicit continuous voice representation. Following a “think-before-speaking” paradigm, a shared language-model backbone first interprets instructions to generate the voice representation, then synthesizes speech conditioned on it. Dual supervision from speaker identity and audio VAE features ensures both identity fidelity and acoustic richness. On InstructTTSEval [16], VoTSpeech achieves leading instruction-following accuracy—particularly on role-play (RP) tasks—alongside top-ranking subjective expressiveness and adherence. Our contributions are summarized as follows:

- We propose VoTSpeech, which separates voice design from speech generation within a shared TTS backbone: instruction interpretation first guides the generation of a continuous voice latent, which then conditions text processing and speech synthesis.
- We construct an automated instruction annotation pipeline with consistency-based quality control to curate high-quality speech–description training pairs.
- We demonstrate superior performance in instruction accuracy, expressiveness, and adherence across benchmark comparisons with modest adaptation data.

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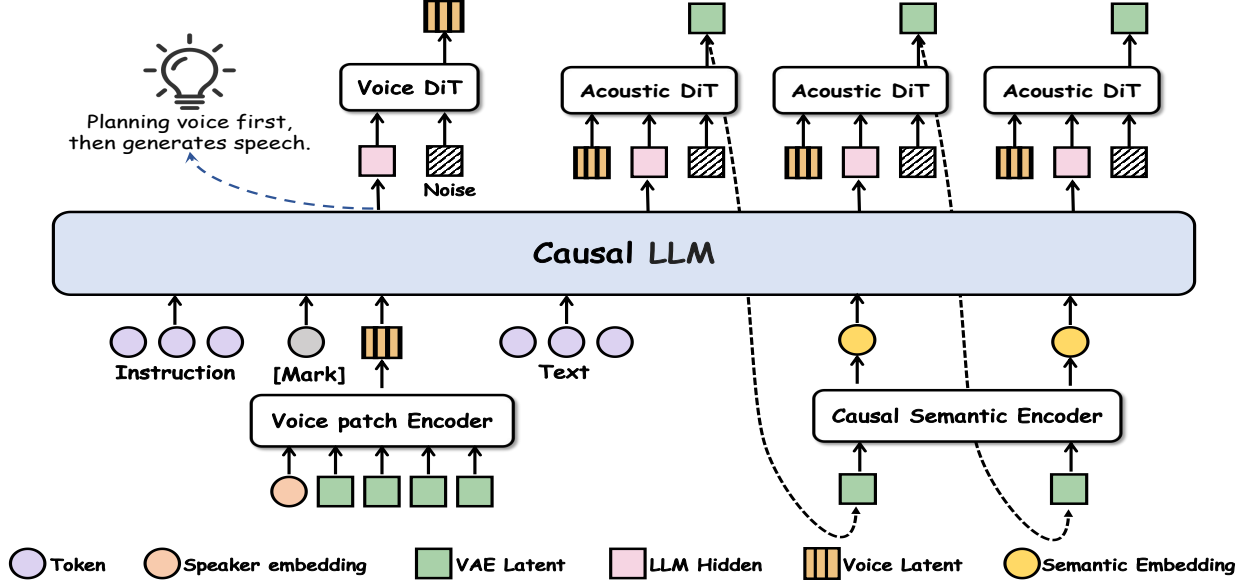


Fig. 1. Overview of VoTSpeech architecture, where an instruction-conditioned Voice DiT predicts a continuous voice latent to condition the language model and Acoustic DiT, jointly supervised by speaker identity and audio VAE features.

2. METHOD

VoTSpeech decouples voice design from speech generation via an explicit continuous voice latent. As shown in Fig. 1, VoTSpeech employs a single shared causal large language model (Causal LLM) backbone for both tasks. This unified design harnesses the Causal LLM’s contextual understanding to interpret text instructions and guide voice latent generation through a Voice Diffusion Transformer (Voice DiT), before synthesizing speech conditioned on the designed voice. VoTSpeech retains the Causal Semantic Encoder and Acoustic DiT from dots.tts [17], while introducing the Voice Patch Encoder and Voice DiT for explicit voice modeling.

2.1. Instruction Data Construction

To distinguish persistent voice identity from dynamic speech delivery, we construct instruction–speech pairs across two complementary annotation levels:

Speaker-Level Inherent Trait Annotation: Multiple recordings from the same speaker are grouped and processed together to capture persistent attributes (e.g., perceived age, gender, pitch register, vocal texture, and accent), following ParaSpeechCaps [18]. The resulting verified description is assigned to all utterances belonging to that speaker.

Utterance-Level Dynamic Style Annotation: Individual recordings are annotated as free-form, actionable control instructions detailing localized delivery dynamics. Rather than relying on rigid templates, these descriptions highlight salient acoustic attributes, including pitch range, stress and volume placement, tempo progression, and emotional trajectory.

PromptDialog and Genshin recordings provide source data for these annotations, complemented by rewritten tag descriptions from ISCSLP2026-CoT-TTS [19]. To improve annotation quality and reduce semantic drift in LLM-generated text [20], Qwen3-Omni [21] directly processes the audio inputs (multi-utterance groups for speaker-level and single recordings for utterance-level) to sample two candidate descriptions per instance. DeepSeek-V4 [22] then evaluates attribute-level consistency between candidates: instances with major contradictions are discarded, whereas localized conflicts are resolved by pruning ambiguous attributes and retaining consensus content.

2.2. Model Architecture

Sequence organization. VoTSpeech places voice design before synthesis text processing while retaining the backbone’s original voice-before-text conditioning order:

$$[c, \langle v \rangle, e_v, x, \ell_{1:T}], \quad (1)$$

where c denotes the voice instruction, $\langle v \rangle$ the voice-generation marker, e_v the continuous voice embedding, x the synthesis text, and $\ell_{1:T}$ the speech-latent sequence. Under causal attention, the hidden state at $\langle v \rangle$ summarizes the instruction prefix without accessing the subsequent synthesis text and conditions voice latent generation. The resulting latent is mapped to e_v and inserted immediately before x , matching the placement of speaker conditioning in the pretrained backbone. This organization introduces an explicit voice-design stage while retaining the familiar conditioning order for syn-

thesis, with the instruction context remaining accessible for expressive control.

Voice Patch Encoder. Prior voice conversion studies show that overly restrictive bottlenecks can trade reconstruction quality for stronger disentanglement [23, 24]. To provide acoustic information for downstream synthesis without extra disentanglement objectives, the Voice Patch Encoder constructs target voice latent z^* by pairing speaker identity with global acoustic cues. Specifically, z^* is constructed from features extracted from waveform y :

$$z^* = \text{norm}([E_{\text{spk}}(y); \text{pool}(E_{\text{vae}}(y))]), \quad (2)$$

where E_{spk} is a pretrained CAM++ encoder [25], E_{vae} is the backbone’s audio VAE encoder, pool denotes temporal mean-pooling over valid frames, and norm applies per-dimension standardization using dataset statistics. Conditioned on h_c , the flow-matching voice prior, Voice DiT, generates voice latent z from Gaussian noise using z^* as its supervision target.

Dual-path voice conditioning. Effective voice control requires aligning high-level prosodic planning in language modeling with low-level timbre rendering in acoustic synthesis. Thus, the generated voice latent z concurrently conditions both Causal LLM processing and acoustic generation:

$$\begin{aligned} e_v &= P_z(z) + P_h(h_c), \\ g_v &= G_s(\tilde{s}) + G_a(\tilde{a}) + G_h(h_c), \end{aligned} \quad (3)$$

where $[\tilde{s}; \tilde{a}] = \text{norm}^{-1}(z)$ recovers speaker and acoustic components, and P, G are learned projections. Here, e_v is prepended to synthesis text x to condition LLM states, fed back via a Causal Semantic Encoder. Combined with g_v , these states guide Acoustic DiT to generate speech latents for VAE decoding. Zero-initialized instruction residuals $P_h(h_c)$ and $G_h(h_c)$ retain instruction nuances beyond z .

2.3. Training and Inference

Joint Optimization and Gradient Decoupling. We train the Voice DiT and speech generation model jointly while keeping E_{spk} and E_{vae} frozen. The Voice DiT learns a conditional vector field between Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ and the target voice latent z^* . For $t \sim \mathcal{U}(0, 1)$ and interpolated state $z_t = (1 - t)\epsilon + tz^*$, the voice prior objective is:

$$\mathcal{L}_{\text{voice}} = E \left[\|v_\theta(z_t, t, \text{sg}(h_c)) - (z^* - \epsilon)\|_w^2 \right], \quad (4)$$

where the expectation is over training pairs, noise, and time, and $\text{sg}(\cdot)$ denotes the stop-gradient operator. The weighted squared norm averages errors within the speaker and acoustic feature slices separately, with respective weights of 1.0 and 0.25. The operation $\text{sg}(h_c)$ prevents the voice-prior loss from updating the Causal LLM, while speech-generation gradients remain active through the conditioning paths.

To mitigate the mismatch between training targets z^* and inference predictions z , speech generation adopts a scheduled sampling strategy that progressively increases the probability of conditioning on generated voice latents. The same selected latent is used in both conditioning paths, without backpropagation through sampling. The overall objective is $\mathcal{L} = \mathcal{L}_{\text{speech}} + \mathcal{L}_{\text{voice}}$, where $\mathcal{L}_{\text{speech}}$ retains acoustic generation and sequence termination losses.

Low-Overhead Inference Execution. VoTSpeech executes a two-stage prefill strategy with prefix KV-cache reuse. First, in Voice Prior Sampling, the Causal LLM processes prefix $[c, \langle v \rangle]$ to extract h_c , enabling Voice DiT to sample latent z —acting like a brief “thought pause” to establish the voice tone. Second, in Resumed Autoregressive Synthesis, after inserting e_v , the model resumes prefill over text x using cached KV states before autoregressively generating speech latents. This avoids redundant instruction encoding with a single sampling step per request, consistently reusing the sampled latent across all subsequent autoregressive decoding steps.

3. EXPERIMENTS

3.1. Experimental Setup

Baselines and Ablations. We compare VoTSpeech against competitive instruction-guided baselines: Ming-Omni-TTS [4], Qwen3-TTS-12Hz-1.7B-VD [5], MOSS-VoiceGenerator [6], and VoxCPM2 [7]. Additionally, as the raw *dots.ts* backbone [17] lacks inherent instruction capabilities, we establish a directly fine-tuned variant (**FT**) trained on the same data. FT retains the backbone’s native conditioning—feeding h_c directly into the acoustic DiT—while omitting the explicit voice prior and dual-path conditioning, serving as a strict baseline for controlled comparison.

Dataset and Implementation Details. Models are trained on the Chinese corpus presented in Sec. 2.1, comprising 1,543.52 hours of unique audio recordings (each audio file is counted once regardless of multiple instruction annotations). VoTSpeech is initialized from *dots.ts* [17] and trained on $8 \times \text{NVIDIA A100 GPUs}$ for 20k steps (≈ 17 epochs). Initial learning rates are set to 10^{-5} for the backbone LLM and 10^{-4} for the voice-design modules. The Voice DiT features a 4-layer Transformer architecture with a hidden dimension of 512 and 8 attention heads. For scheduled sampling during synthesis training, the probability of conditioning on generated voice latents is 0 for the first 6k steps, increases linearly to 0.75 over the subsequent 3k steps, and remains fixed at 0.75 thereafter. Voice latent sampling uses 2 integration steps during training and 16 at inference.

Evaluation Metrics. Following the InstructTTEval-ZH protocol [16], we employ Gemini 2.5 Pro [26] to evaluate instruction compliance, reporting accuracy (%) across Acoustic-Parameter Specification (APS), Descriptive-Style Directive (DSD), and Role-Play (RP). Intelligibility is mea-

Table 1. Comparison with existing systems. Bold and underline mark the best and second-best means.

Model	Objective Evaluation				Subjective Evaluation		
	APS	DSD	RP	CER ↓	NMOS	EMOS	AMOS
Ming-Omni-TTS-0.5B	<u>84.9</u>	72.2	53.9	2.48	3.71 ± 0.27	3.71 ± 0.24	3.75 ± 0.25
Qwen3-TTS-12Hz-1.7B-VD	87.1	<u>76.0</u>	55.2	2.99	<u>4.08 ± 0.21</u>	<u>4.10 ± 0.19</u>	<u>4.06 ± 0.20</u>
MOSS-VoiceGenerator	73.1	70.2	<u>58.9</u>	5.03	3.86 ± 0.22	3.72 ± 0.18	3.70 ± 0.19
VoxCPM2	84.7	71.8	56.8	<u>2.58</u>	3.80 ± 0.25	3.82 ± 0.18	3.82 ± 0.17
VoTSpeech (Ours)	84.8	77.3	66.5	2.58	4.12 ± 0.22	4.30 ± 0.18	4.26 ± 0.19

sured via overall character error rate (CER) across all categories using a pretrained Paraformer-ZH ASR model [27]. For subjective evaluation, 24 listeners rate outputs from each system on the same 10 randomly selected evaluation examples using a five-point scale. We report mean opinion scores for naturalness (NMOS), expressiveness (EMOS), and instruction adherence (AMOS), assessing perceived audio quality, how well the vocal delivery fits the synthesis text, and compliance with voice-design instructions, respectively. Subjective scores are reported with 95% confidence intervals (CIs) in Table 1. Higher values indicate better performance for all metrics except CER.

3.2. Experimental Results

Main Results. Table 1 compares VoTSpeech with existing instruction-guided TTS systems. VoTSpeech achieves the highest accuracy on DSD at 77.3% and RP at 66.5%, exceeding top baselines by 1.3 and 7.6 percentage points, respectively. On APS, its accuracy of 84.8% trails Qwen3-TTS by 2.3 percentage points while remaining competitive with other baselines. These results show a clear advantage in handling complex descriptive and role-play instructions. Meanwhile, VoTSpeech preserves high speech intelligibility with a competitive CER of 2.58%. Subjective evaluations complement these objective findings: VoTSpeech earns the highest mean ratings across all dimensions, scoring 4.253 for naturalness, 4.420 for expressiveness, and 4.317 for instruction adherence. Overall, these findings validate explicit voice modeling for controllable speech synthesis.

Ablation Study. Table 2 evaluates the contribution of explicit voice modeling by comparing the full VoTSpeech framework against direct backbone fine-tuning. Incorporating the intermediate voice representation improves DSD and RP accuracy by 2.3 and 2.2 percentage points, respectively, while reducing CER from 2.71% to 2.58%. All subjective metrics show consistent gains, with instruction adherence achieving the largest improvement (+0.300). This aligns closely with the objective gains on DSD and RP, confirming that predicting an intermediate voice prior effectively bridges natural-language directives and acoustic generation. In contrast, APS shows only a marginal shift (−0.5 percentage points), indicating that the primary benefit of explicit voice

Table 2. Ablation against direct fine-tuning. Δ denotes Ours minus FT.

Model	APS	DSD	RP	CER↓	NMOS	EMOS	AMOS
FT	85.30	75.00	64.30	2.71	3.87	4.12	3.96
Ours	84.80	77.30	66.50	2.58	4.12	4.30	4.26
Δ	−0.50	+2.30	+2.20	−0.13	+0.25	+0.19	+0.30

modeling lies in capturing holistic voice characteristics and style rather than isolated acoustic parameters.

Discussion. The empirical results illuminate both the strengths and intrinsic bounds of our voice prior. By design, VoTSpeech condenses user instructions into a single continuous token supervised by speaker and pooled acoustic features. While compact, this global latent effectively captures macro-level stylistic directives and complex persona nuances (DSD and RP), bridging semantics and acoustics. Conversely, its performance on explicit acoustic specifications shows that a static global representation inherently struggles with localized or time-varying control (e.g., pitch or tempo variations). This reveals a fundamental trade-off: a compact global prior provides a strong inductive bias for holistic voice timbre, whereas fine-grained attribute control requires multi-scale or temporally dense representations. Unifying global priors with localized control presents a promising direction for future research.

4. CONCLUSION

We introduced VoTSpeech to bridge natural-language directives and speech generation by explicitly modeling an intermediate voice prior. Supervised by multi-level acoustic annotations, our flow-based prior decouples macro-level voice design from fine-grained acoustic decoding. Comprehensive evaluations demonstrate that this separation significantly improves adherence to complex descriptive and role-play instructions while preserving speech quality and intelligibility. Ultimately, our findings confirm explicit voice modeling as a viable paradigm for controllable speech synthesis.

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